

Eng Math - II (Probability Theory)

Practice Problem Set - 4

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1. (a) How many different 7 place license plates are possible if the first 2 places are for letters and the other 5 for numbers ?
(b) Repeat part (a) under the assumption that no letter or number can be repeated in a single license plate.
2. How many different 7 place license plates are possible when 3 of the entries are letters and 4 are digits ? Assume that repetition of letters and numbers is allowed and that there is no restriction on where the letters or numbers can be placed.
3. Twenty workers are to be assigned to 20 different jobs, one to each job. How many different assignments are possible ?
4. (a) In how many ways can 3 boys and 3 girls sit in a row ?
(b) In how many ways can 3 boys and 3 girls sit in a row if the boys and the girls are each to sit together ?
(c) In how many ways if only the boys must sit together ?
(d) In how many ways if no two people of the same sex are allowed to sit together ?
5. How many different letter arrangements can be made from the letters:
(a) Fluke;
(b) Propose;
(c) Mississippi;
(d) Arrange ?
6. How many different linear arrangements are there of the letters A, B, C, D, E and F for which
(a) A and B are next to each other;
(b) A is before B;
(c) A is before B and B is before C;
(d) A is before B and C is before D;
(e) A and B are next to each other and C and D are also next to each other;
(f) E is not last in line ?
7. If 4 Americans, 3 French people, and 3 British people are to be seated in a row, how many seating arrangements are possible when people of the same nationality must sit next to each other ?
8. A child has 12 blocks, of which 6 are black, 4 are red, 1 is white, and 1 is blue. If the child puts the blocks in a line, how many arrangements are possible ?

9. In how many ways can 8 people be seated in a row if
 - (a) there are no restrictions on the seating arrangement;
 - (b) persons A and B must sit next to each other;
 - (c) there are 4 men and 4 women and no 2 men or 2 women can sit next to each other;
 - (d) there are 5 men and they must sit next to one another;
 - (e) there are 4 married couples and each couple must sit together ?
10. In how many ways can 3 novels, 2 mathematics books, and 1 chemistry book be arranged on a bookshelf if
 - (a) the books can be arranged in any order;
 - (b) the mathematics books must be together and the novels must be together.
 - (c) The novels must be together, but the other books can be arranged in any order ?
11. Consider a group of 20 people. If everyone shakes hands with everyone else, how many handshakes take place ?
12. How many 5 card poker hands are there ?
13. From a group of 8 women and 6 men, a committee consisting of 3 men and 3 women is to be formed. How many different committees are possible if
 - (a) 2 of the men refuse to serve together;
 - (b) 2 of the women refuse to serve together;
 - (c) 1 man and 1 woman refuse to serve together ?
14. If 12 people are to be divided into 3 committees of respective sizes 3, 4, and 5. How many divisions are possible ?
15. If 8 new teachers are to be divided among 4 schools, how many divisions are possible ? What if each school must receive 2 teachers ?
16. How many subsets of size 4 of the set $S = \{1, 2, 3, \dots, 20\}$ contain at least one of the elements 1, 2, 3, 4, 5 ?
17. Consider n - digit numbers where each digit is one of the 10 integers 0, 1, 2, $\dots$ 9. How many such numbers are there for which
 - (a) no two consecutive digits are equal;
 - (b) 0 appears as a digit a total of i times, $i = 0, 1, \dots, n$?
18. A student has to sell 2 books from a collection of 6 math, 7 science, and 4 economics books. How many choices are possible if
 - (a) both books are to be on the same subject;
 - (b) the books are to be on different subjects ?
19. How many terms are there in the multinomial expansion of $(x_1 + x_2 + \dots + x_n)^n$.
20. If 8 identical blackboards are to be divided among 4 schools. How many divisions are possible ? How many, if each school must receive at least 1 blackboard ?
21. If two dice are rolled, what is the probability that the sum of the upturned faces will equal 9 ?

22. If 4 balls are “randomly drawn” from a bowl containing 8 white and 6 black balls, what is the probability that one of the balls is white and the other three black ?
23. A committee of 7 is to be selected from a group of 8 men and 9 women. If the selection is made randomly, what is the probability that the committee consists of 5 men and 2 women ?
24. An urn contains 100 balls, one of which is special. If 10 of these balls are withdrawn one at a time, with each selection being equally likely to be any of the balls that remain at the time, what is the probability that the special ball is chosen ?
25. If 23 people are present in a room, what is the probability that no two of them celebrate their birthday on the same day of the year ?
26. A box contains 3 marbles: 1 red, 1 green, and 1 blue. Consider an experiment that consists of taking 1 marble from the box and then replacing it in the box and drawing a second marble from the box. Describe the sample space. Repeat when the second marble is drawn without replacing the first marble.
27. Two dice are thrown. Let E be the event that the sum of the dice is odd, let F be the event that at least one of the dice lands on 1, and let G be the event that the sum is 5. Describe the events $E \cap F$, $E \cup F$, $F \cap G$, $E \cap F^c$, and $E \cap F \cap G$.
28. Suppose that A and B are mutually exclusive events for which $P(A) = .3$ and $P(B) = .5$. What is the probability that:
 - (a) either A or B occurs ?
 - (b) A occurs but B does not ?
 - (c) both A and B occur ?
29. An elementary school is offering 3 language classes: one in Spanish, one in French, and one in German. The classes are open to any of the 100 students in the school. There are 28 students in the Spanish class, 26 in the French class. and 16 in the German class. There are 12 students who are in both Spanish and French, 4 who are in both Spanish and German, and 6 who are in both French and German. In addition, there are 2 students taking all 3 classes.
 - (a) If a student is chosen randomly, what is the probability that he or she is not in any of the language classes ?
 - (b) If a student is chosen randomly, what is the probability that he or she is taking exactly one language class ?
 - (c) If 2 students are chosen randomly, what is the probability that at least 1 is taking a language class ?
30. Poker dice is played by simultaneously rolling 5 dice. Show that:
 - (a) $P(\text{no two alike}) = .0926$
 - (b) $P(\text{one pair}) = .4630$
 - (c) $P(\text{two pair}) = .2315$
 - (d) $P(\text{three alike}) = .1543$
 - (e) $P(\text{full house}) = .0386$

(f) $P(\text{four alike}) = .0193$

(g) $P(\text{five alike}) = .0008.$

31. An urn contains 5 red, 6 blue and 8 green balls. If a set of 3 balls is randomly selected, what is the probability that each of the balls will be :
- (a) of the same color ?
 - (b) of different colors ?
32. Repeat the previous problem 11, under the assumption that whenever a ball is selected, its color is noted and it is then replaced in the urn before the next selection. This is known as sampling *with replacement*.
33. An urn contains n white and m black balls, where n and m are positive numbers.
- (a) If two balls are randomly withdrawn, what is the probability that they are the same color ?
 - (b) If a ball is randomly withdrawn and then replaced before the second one is drawn, what is the probability that the withdrawn balls are the same color ?
 - (c) Show that the probability in part (b) is always larger than the one in part (a).
34. A pair of fair dice is rolled. What is the probability that the second die lands on a higher value than does the first ?
35. If two dice are rolled, what is the probability that the sum of the upturned faces equals i ? Find it for $i = 2, 3, \dots, 11, 12$.
36. Seven balls are randomly withdrawn from an urn that contains 12 red, 16 blue and 18 green balls. Find the probability that:
- (a) 3 red, 2 blue and 2 green balls are withdrawn
 - (b) at least 2 red balls are withdrawn
 - (c) all withdrawn balls are the same color
 - (d) either exactly 3 red balls or exactly 3 blue balls are withdrawn.
37. Five people, designated as A, B, C, D, E are arranged in linear order. Assuming that each possible order is equally likely, what is the probability that:
- (a) there is exactly one person between A and B
 - (b) there are exactly two people between A and B
 - (c) there are three people between A and B ?
38. From a group of 3 first-year students, 4 sophomores, 4 juniors and 3 seniors, a committee of size 4 is randomly selected. Find the probability that the committee will consist:
- (a) 1 from each class
 - (b) 2 sophomores and 2 juniors
 - (c) only sophomores or juniors.
39. A six digit number is chosen at random. What is the probability that at least two consecutive digits in the chosen number are equal ?
40. Urmi is 80 percent certain that his missing key is in one of the two pockets of his hanging jacket, being 40 percent certain it is in the left-hand pocket and 40 percent certain it is in the right-hand pocket. If a search of the left-hand pocket does not find the key, what is the conditional probability that it is in the other pocket ?

41. A coin is flipped twice. Assuming that all four points in the sample space

$$\mathbb{S} = \{(H, H), (H, T), (T, H), (T, T)\}$$

are equally likely.

What is the conditional probability that both flips land on heads, given that:

(a) the first flip lands on heads,

(b) at least one flip lands on heads ?

42. In the card game bridge, the 52 cards are dealt out equally to 4 players—called East, West, North and South. If North and South have a total of 8 spades among them. What is the probability that East has 3 of the remaining 5 spades ?

43. Suppose that an urn contains 8 red balls and 4 white balls. We draw 2 balls from the urn without replacement. If we assume that at each draw, each ball in the urn is equally likely to be chosen, what is the probability that both balls drawn are red ?

44. An ordinary deck of 52 playing cards is randomly divided into 4 piles of 13 cards each. Compute the probability that each pile has exactly 1 ace.

45. Two fair dice are rolled. What is the conditional probability that at least one lands on 6 given that the dice land on different numbers ?

46. If two fair dice are rolled, what is the conditional probability that the first one lands on 6 given that the sum of the dice is i ? Compute for all values of i between 2 and 12.

47. Compute in the hand of bridge the conditional probability that East has 3 spades given that North and South have a combined total of 8 spades.

48. What is the probability that at least one of a pair of fair dice lands on 6, given that the sum of the dice is i , $i = 2, 3, \dots, 11, 12$.

49. An urn contains 6 white and 9 black balls. If 4 balls are to be randomly selected without replacement, what is the probability that the first 2 selected are white and the last 2 black ?

50. The king comes from a family of 2 children. What is the probability that the other child is his sister ?

51. A couple has 2 children. What is the probability that both are girls if the older of the two is a girl ?

52. An urn initially contains 5 white and 7 black balls. Each time a ball is selected, its colour is noted and it is replaced in the urn along with 2 other balls of the same colour. Compute the probability that:

(a) the first 2 balls selected are black and the next 2 are white,

(b) of the first 4 balls selected, exactly 2 are black ?

53. In a certain community, 36 percent of the families own a dog and 22 percent of the families that own a dog also own a cat. In addition, 30 percent of the families own a cat. What is:

(a) the probability that a randomly selected family owns both a dog and a cat ?

(b) the conditional probability that a randomly selected family owns a dog given that it owns a cat ?

54. There are 3 coins in a box. One is a two-headed coin, another is a fair coin, and the third is a biased coin that comes up heads 75 percent of the time. When one of the 3 coins is selected at random and flipped, it shows heads. What is the probability that it was the two-headed coin ?

55. Show that if $P(A/B) = 1$, then $P(B^c/A^c) = 1$.

56. There is a 60 percent chance that event A will occur. If A does not occur, then there is a 10 percent chance that B will occur. What is the probability that at least one of the events A or B will occur ?

57. Show that if $P(A) > 0$, then

$$P(A \cap B/A) \geq P(A \cap B/A \cup B).$$

58. Let $A \subset B$. Express the following probabilities as simply as possible:

$$P(A/B), P(A/B^c), P(B/A), P(B/A^c).$$

59. A family has n children with probability p_n , where $p_1 = .1$, $p_2 = .25$, $p_3 = .35$, $p_4 = .3$. A child from this family is randomly chosen. Given that this child is the eldest child in the family, find the conditional probability that the family has:

(a) only 1 child,

(b) 4 children.

Redo (a) and (b) when the randomly selected child is the youngest child of the family.

60. Suppose that we toss 2 fair dice. Let E denote the event that the sum of the dice is 6 and F denote the event that the first die equals 4. Examine whether E and F are independent.

61. Multiple-choice test consists of 20 questions. Each question has four choices, exactly one of which is correct. For each question, a student is able to correctly identify one of the choices as wrong and chooses one answer at random from the remaining three choices. The student will get a scholarship if she answers at least 18 questions correctly. What is the probability that she gets the scholarship ?

62. Two balls are chosen randomly from an urn containing 8 white, 4 black, and 2 orange balls. Suppose that we win 2 rupees for each black ball selected and we lose 1 rupees for each white ball selected. Let X denote our winnings. What are the possible values of X , and what are the probabilities associated with each value ?

63. Two fair dice are rolled. Let X equal the product of the 2 dice. Compute $P(X = i)$ for $i = 1, \dots, 36$.

64. Three dice are rolled. Find the probabilities attached to the possible values that X can take on, where X is the sum of the 3 dice.

65. 5 men and 5 women are ranked according to their scores on an examination. Assume that no two scores are alike and all $10!$ possible rankings are equally likely. Let X denote the highest

ranking achieved by a woman (For instance, $X = 1$ if the top-ranked person is female).

Find $P(X = i)$, $i = 1, 2, 3, \dots, 8, 9, 10$.

66. Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. What are the possible values of X ?. For $n = 3$, if the coin is assumed fair, what are the probabilities associated with the values that X can take on ?

67. Suppose that a die is rolled twice. What are the possible values that the following random variables can take on:

- (a) the maximum value to appear in the two rolls,
- (b) the minimum value to appear in the two rolls,
- (c) the sum of the two rolls,
- (d) the value of the first roll minus the value of the second roll ?.

Find the probabilities associated with the random variables in parts (a) through (d).

68. Four balls are to be randomly selected with replacement from an urn that contains 20 balls numbered 1 through 20. If X is the largest numbered ball selected. What are the possible values that X can take on. Also find the value of $P(X > 10)$.

69. Plot the graph of the pmf of the random variable representing the sum when two dice are rolled.

70. Let X be the winnings of a gambler and $p(i) = P(X = i)$ be pmf of X . Assume that

$$p(0) = \frac{1}{3}, \quad p(1) = p(-1) = \frac{13}{55}, \quad p(2) = p(-2) = \frac{1}{11}, \quad p(3) = p(-3) = \frac{1}{165}.$$

Find the conditional probability that the gambler wins i , $i = 1, 2, 3$ given that he/she wins a positive amount.

71. Four independent flips of a fair coin are made. Let X denote the number of heads obtained. Plot the pmf of the random variable $X - 2$.

72. If the distribution function of X is given by

$$F(b) = \begin{cases} 0 & b < 0 \\ \frac{1}{2} & 0 \leq b < 1 \\ \frac{3}{5} & 1 \leq b < 2 \\ \frac{4}{5} & 2 \leq b < 3 \\ \frac{9}{10} & 3 \leq b < 3.5 \\ 1 & b \geq 3.5. \end{cases}$$

Find the probability mass function of X .

73. Suppose that the distribution function of X is given by

$$F(b) = \begin{cases} 0 & b < 0 \\ \frac{b}{4} & 0 \leq b < 1 \\ \frac{1}{2} + \frac{b-1}{4} & 1 \leq b < 2 \\ \frac{11}{12} & 2 \leq b < 3 \\ 1 & b \geq 3. \end{cases}$$

(a) Find $P(X = i)$ for $i = 1, 2, 3$ and $P(\frac{1}{2} < X < \frac{3}{2})$.

74. Suppose that two teams play a series of games that ends when one of them has won i games. Suppose that each game played is independently won by team A with probability p . Find the expected number of games that are played when (a) $i = 2$ and (b) $i = 3$. Also, show in both cases that this number is maximised when $p = \frac{1}{2}$.

75. If $E[X] = 1$ and $\text{Var}(X) = 5$ then find:

(a) $E[(2 + X)^2]$,

(b) $\text{Var}(4 + 3X)$.

76. When coin 1 is flipped, it lands on heads with probability .4. When coin 2 is flipped, it lands on heads with probability .7. One of these coins is randomly chosen and flipped 10 times.

(a) What is the probability that the coin lands on heads on exactly 7 of the 10 flips ?

(b) Given that the first of these 10 flips lands heads, what is the conditional probability that exactly 7 of the 10 flips land on heads ?

77. Suppose that a biased coin that lands on heads with probability p is flipped 10 times. Given that a total of 6 heads results, find the conditional probability that the first 3 outcomes are:

(a) h, t, t (means the first flip results in heads, the second in tails, and the third in tails)

(b) t, h, t .

78. Compare the Poisson approximation with the correct binomial probability for the following cases:

(a) $P(X = 2)$ when $n = 8$, $p = .1$,

(b) $P(X = 9)$ when $n = 10$, $p = .95$,

(c) $P(X = 0)$ when $n = 10$, $p = .1$,

(d) $P(X = 4)$ when $n = 9$, $p = .2$.

79. If you buy a lottery ticket in 50 lotteries, in each of which your chance of winning a prize is $\frac{1}{100}$, what is the (approximate) probability that you will win a prize:
- (a) at least once ?
 - (b) exactly once ?
 - (c) at least twice ?

80. A fair coin is continually flipped until heads appear for the 10th time. Let X denote the number of tails that occur. Compute the probability mass function of X .

81. Let X be random variable such that

$$P(X = 1) = p = 1 - P(X = -1).$$

Find $c \neq 1$ such that $E[c^X] = 1$.

82. Find $\text{Var}(X)$ if

$$P(X = a) = p = 1 - P(X = b).$$

83. If X is a binomial random variable such that $E[X] = 6$ and $\text{Var}(X) = 2.4$. Find $P(X = 5)$.